

*“The limits of computable functions lie not in nature,
but in our definitions.”*

— Alan Turing, 1936

The Arithmetic of Order: A Computable Foundation for Reality

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Dedication

*This work is dedicated to **Alan Mathison Turing** (1912–1954),
whose pioneering insights into the limits of logic, the nature of computation,
and the generative power of discrete processes laid the foundation for
a new understanding of mathematics, physics, and intelligence.*

*The Arithmetic of Order (AoO) seeks to extend Turing’s vision:
that finite operations, governed by internal logic,
can unfold into the full richness of the physical world.*

Abstract

This work presents the Arithmetic of Order (AoO), a framework for the foundations of physics built upon a single, finitistic principle. Rejecting the continuum as axiomatic, AoO demonstrates the emergence of physical reality from a discrete, computable, and non-associative algebraic substrate. We formalize the *Origin–Observation Duality*, where a single tri-octonion state vector v generates a unique Hermitian Jordan operator A that satisfies both a nonlinear internal eigen-equation ($Av = v[v]$) and admits a classical real-valued spectral decomposition ($Aw = \lambda w$). This mechanism provides a deductive origin for an emergent 3+1D spacetime and a deterministic model for particle interactions. We reformulate LHC jet production as the spectral footprint of a G_2 automorphism cascade, a non-perturbative process governed by the internal logic of octonion algebra. Finally, we show how the core principles of AoO are mirrored in the architectures of modern generative models for particle physics, grounding the theory in experimental and computational practice.

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Introduction

Historical Context. The Arithmetic of Order (AoO) draws deeply on the legacy of foundational thinkers—from Hamilton and Cayley to Jordan, Turing, and Conway. It posits that the universe is not described by mathematics but is, in its deepest sense, a mathematical structure emerging from finite and computable rules [1, 3]. This paper aims to formalize the core algebraic engine of this framework, building upon the conceptual foundations laid out in [5]. We will demonstrate how a single, computable principle—the Origin-Observation Duality—gives rise to the structure of physical law and connects directly to the frontiers of experimental and computational physics.

1 Review of Supporting and Analogous Works

1.1 Conceptual Foundations in the Arithmetic of Order

The present work is a formal, mathematical extension of the principles first articulated in "The Arithmetic of Order: A Finitistic and Topological Foundation for Reality" [5]. That foundational draft introduced the core conceptual pillars that this paper now grounds in rigorous algebraic machinery:

- **Topological Particles:** The idea that fundamental particles are not points but topological objects, such as braids and knots.
- **The Associator Projection:** The principle that the non-associative structure of the octonions is the "engine" of reality, which must project down to a stable, observable form. This is the conceptual precursor to the $Av = v[v]$ equation.
- **Jets as Topological Shockwaves:** The reinterpretation of collider jets as deterministic, topological events rather than statistical, probabilistic sprays. This intuition is now formalized via the G_2 automorphism cascade.

This paper provides the specific algebraic mechanism—the Origin-Observation Duality—that realizes these foundational concepts.

1.2 Computational Analogues of Generative Logic

Recent advances in generative AI for particle physics provide a striking computational reflection of the AoO's generative principles. The paper "Extrapolating Jet Radiation with Autoregressive Transformers" (JetGPT) [9] is a key example.

- **Factorized Generation:** The transformer learns to generate the next particle in a sequence based on the history of previous particles, $p(x_i|x_{1:i-1})$. This probabilistic, autoregressive grammar is a direct computational analogue to the AoO's deterministic, sequential G_2 automorphism cascade, where the state v_{n+1} is an algebraic function of v_n .
- **Learned Universality:** The ability of JetGPT to extrapolate to higher jet multiplicities suggests it has learned a "universal splitting kernel." In AoO, this universality is not statistical but algebraic—it is the fixed, deterministic logic of the octonion algebra's automorphism group.

1.3 Computational Analogues of Observation and Unfolding

The challenge of inverting detector measurements to infer the underlying "truth" provides a powerful real-world example of the AoO's observational projection. The "Simulation-Prior Independent Neural Unfolding Procedure" (SPINUP) [10] is particularly relevant.

- **The Unfolding Problem as the Duality Inverse:** SPINUP's goal is to solve for $p(\text{truth}|\text{observation})$. This is precisely the inverse of the AoO's "Great Projection": inferring the internal state v from the observable spectrum λ .
- **Information Loss and Smoothing:** The SPINUP paper notes the inherent information loss in the forward process and the tendency of neural networks to learn "smoother, less peaked functions." This is a direct experimental signature of what AoO predicts: when a fundamentally discrete, quantized algebraic reality ($Av = v[v]$) is projected into an observable, continuous measurement model ($Aw = \lambda w$), the sharp, discrete features are inevitably smoothed.

2 The Core Algebraic Engine

2.1 The Tri-Octonion State and the Associator

The fundamental unit of information in AoO is a **tri-octonion state vector**:

$$v = (o_1, o_2, o_3) \in \text{Im}(\mathbb{O})^3$$

Its intrinsic topological structure is quantified by the **associator**:

$$[v] := (o_1 o_2) o_3 - o_1 (o_2 o_3) \in \text{Im}(\mathbb{O})$$

This non-zero "non-associative charge" is the engine of all dynamics.

2.2 The Origin–Observation Duality

The central mechanism of AoO is a duality that separates the internal, generative logic of a system from its external, measurable manifestation.

Proposition 2.1 (Origin–Observation Duality). *Let $v \in \text{Im}(\mathbb{O})^3$ be a tri-octonion state with associator $[v]$. Let $A \in \text{Herm}_3(\mathbb{O})$ be the unique Hermitian Jordan operator constructed from v and six real parameters. The following duality holds:*

1. **Origin of Law (Internal Logic):** *The operator A is selected by the nonlinear, self-referential eigen-equation:*

$$Av = v[v]$$

This equation defines the internal, unobservable law consistent with the state's algebraic topology. The associator $[v]$ acts as a hidden, non-real eigenvalue.

2. **Observation of Law (External Measurement):** *The same operator A , as an element of a formally real Jordan algebra, has a classical spectral decomposition:*

$$Aw = \lambda w, \quad \lambda \in \mathbb{R}, \quad w \in \mathbb{O}^3,$$

where the eigenvalues $\{\lambda_i\}$ are real and the eigenvectors $\{w^{(i)}\}$ are orthogonal. These correspond to the quantized, measurable observables of the system.

Remark 2.2 (Foundational Acknowledgement). The construction of the operator A satisfying the internal eigen-equation is based on the foundational work by Gillow-Wiles and Dray [6]. They proved that any purely imaginary, non-quaternionic vector v admits a 6-parameter family of Hermitian matrices satisfying $Av = v[v]$. In the AoO, this mathematical result is interpreted as a physical law-generation mechanism.

3 Emergent Spacetime and the Turing Emergence Theorem

3.1 The Emergent 3+1D Structure

- **3D Space:** The three real eigenvalues $(\lambda_1, \lambda_2, \lambda_3)$ and eigenvectors $(w^{(1)}, w^{(2)}, w^{(3)})$ of A define a local, dynamic 3D reference frame for observation.
- **1D Time:** The evolution of the system, $v_n \rightarrow v_{n+1}$, is driven by the irreversible, non-associative dynamics of the associator. The sequence $[v_n]$ acts as a discrete, intrinsic algebraic clock.

3.2 The Turing Emergence Theorem

This entire process, from a discrete seed to an observable reality, fulfills the vision of Alan Turing.

Theorem 3.1 (Turing Emergence). *Let a physical system be defined by a finite tri-octonion seed $v \in \text{Im}(\mathbb{O})^3$ with a non-zero associator $[v]$. Let its evolution be governed by a finite set of computable, algebraic update rules. The Arithmetic of Order (AoO) framework dictates that this seed deterministically generates a unique Hermitian Jordan operator $A \in \text{Herm}_3(\mathbb{O})$. The classical spectral decomposition of A yields a set of real, measurable eigenvalues $\{\lambda_i\}$ and a corresponding orthogonal frame of eigenvectors $\{w^{(i)}\}$.*

Thus, a discrete, internal, and computable logic (the "Turing Machine" of the state) deterministically unfolds into an observable, externally consistent physical reality (the "Oracle's" output). This realizes Turing's vision of universal computation manifesting as physical law.

4 Reformulating Hadron Collider Physics

4.1 Algebraic Jet Emission via G_2 Automorphism Cascade

AoO provides a deterministic, non-perturbative model for jet production at the LHC. A jet is the spectral footprint of a discrete algebraic transformation.

1. **Initial State:** A quark is modeled as a tri-octonion state v .
2. **Interaction as Automorphism:** A gluon-like interaction is modeled by the action of a G_2 automorphism, $F(S, T, X)$, which transforms the state $v \rightarrow v'$.
3. **Jet Cascade:** Hadronization is a deterministic cascade of these automorphisms, $v \rightarrow v' \rightarrow v'' \dots$
4. **Observable Jet:** The jet is the collection of the spectral outputs from the sequence of difference operators, $\Delta A_k = A^{(k)} - A^{(k-1)}$.

This model predicts quantized energy spectra and angular distributions for jets, as well as the existence of finite families of jet topologies governed by braid group symmetries (see Appendix).

5 Conclusion and Outlook

The Arithmetic of Order offers a complete, computable, and philosophically coherent foundation for physics. By reinterpreting the algebraic structures of the octonions and exceptional Jordan algebras, it provides a deterministic origin for spacetime, particle states, and their interactions. The framework makes concrete, falsifiable predictions for hadron collider physics and finds deep

resonance with the architectures of modern generative AI. Ultimately, AoO extends the legacy of Alan Turing, proposing a universe that is not merely described by computation but is, in its essence, a computation.

A Appendix: Oracle Automorphisms and Klein Character Varieties

The dynamics of the AoO lattice are regulated by higher-order algebraic structures that can be interpreted as "oracles."

A.1 The Automorphism $F(S, T, X)$ as an Interaction Operator

The map $F(S, T, X) := (TS)^{-1}(T(SX))$ is a G_2 automorphism that acts as a gluon-like operator, transforming the color state of a tri-octonion. This is based on the description of inner automorphisms of the octonions discussed by Baez and others [8].

A.2 Generation of Braid Group Symmetries

Compositions of these order-2 automorphisms generate order-3 symmetries of the form PXP^{-1} , providing an algebraic origin for the braid group B_3 and, consequently, the $SU(3)$ color symmetry of QCD.

A.3 The Klein Cubic Surface as a Manifold of Observables

The space of consistent interaction configurations is constrained by a character variety, the **Klein cubic surface**: $xyz + x^2 + y^2 + z^2 = x + y + z$. The discrete, 7-point braid group orbits on this surface correspond to distinct, quantized families of jet topologies.

B Appendix: Impact Analysis of the Turing Emergence Theorem

The Turing Emergence Theorem represents a significant paradigm shift. This appendix outlines its potential impact on scientific research.

Why: The Fundamental Problems It Solves

- **Origin of Spacetime:** Provides a constructive, computable mechanism for the emergence of a 3+1D spacetime.
- **Origin of Quantization:** Asserts that quantization is a necessary consequence of the discrete, real-valued spectra of Hermitian Jordan operators.
- **The Arrow of Time:** Provides a fundamental, dynamical origin for time's irreversibility, rooting it in non-associative evolution.
- **The Measurement Problem:** Resolves the duality between internal quantum logic and external classical measurement via the "Origin-Observation Duality."
- **The Problem of Infinities:** By building physics from a finitistic foundation, it eliminates the need for renormalization.

How: A Paradigm Shift in Scientific Methodology

- **From Calculus to Algebra:** The language of physics shifts from differential equations to discrete, algebraic update rules.
- **From Probabilistic to Deterministic Models:** Stochastic models are replaced by deterministic, computable G_2 automorphism cascades.
- **Algebra-Aware AI:** AI models can be designed to learn the underlying generative grammar of the AoO, effectively reverse-engineering the universe’s source code.

Who: The Scientific Communities Impacted

- **Theoretical Physicists:** A new, unified foundation for quantum gravity and particle physics.
- **Experimental Particle Physicists:** A new set of concrete, falsifiable predictions to test (e.g., quantized spectra, forbidden zones in jet data).
- **Computer Scientists and Information Theorists:** Evidence that the universe is a computation, inspiring new models of computation.
- **Philosophers of Science:** A computable framework to re-evaluate causality, determinism, and emergence.

When: A Roadmap for Adoption and Validation

The validation of a new foundational framework typically unfolds through sequential phases. However, with the accelerating power of ensemble-based artificial intelligence—especially systems capable of symbolic reasoning and algebraic inference—the Arithmetic of Order (AoO) may progress through these phases far more rapidly than traditional paradigms.

- **Foundational Development and Simulation.** Theoretical refinement of the AoO framework continues in tandem with the development of open-source, algebraically grounded simulation tools. These tools allow researchers and AI agents to explore the dynamics of tri-octonion states, associators, and spectral unfolding.
- **Experimental and Computational Integration.** Collaborative efforts emerge between theorists, experimental physicists, and AI researchers to test AoO’s falsifiable predictions. These include distinct spectral bandings, forbidden zones in collider data, and topological jet classifications. Simultaneously, ensemble AIs trained on AoO principles begin to serve as “inductive companions,” guiding data analysis and unfolding processes with algebraic awareness.
- **Paradigm Integration and Technological Application.** As validation accumulates and simulation fidelity increases, AoO may become a unifying language for physics education, computation, and technology. Its non-associative logic could inspire novel quantum architectures, and its computable grammar might seed the next generation of AI models designed not merely to predict data, but to discover laws.

Rather than fixed timelines, the success of this roadmap hinges on the synergy between computable theory and intelligent agents—where both evolve together. AoO not only benefits from this AI acceleration but also informs its design, potentially making the framework concretely sound in the near future.

References

- [1] F. El Khettabi, *The Arithmetic of Order: A Finitistic Foundation for Mathematics...*, viXra:2505.0059 (2025).
- [2] F. El Khettabi, *A Comprehensive Modern Mathematical Foundation for Hypercomplex Numbers...*, viXra:2505.0054 (2025).
- [3] F. El Khettabi, *The Arithmetic of Order: A Unified Origin for the Standard Model's Gauge Symmetries...*, viXra:2507.0086 (2025).
- [4] F. El Khettabi, *Physics as Quantized Measurement: The Farey Sequence as a Realization of the Arithmetic of Order*, viXra:2507.0068 (2025).
- [5] F. El Khettabi, *The Arithmetic of Order: A Finitistic and Topological Foundation for Reality*, [\[Online Manuscript\]](#), (2025).
- [6] H. Gillow-Wiles and T. Dray, *Finding 3×3 Hermitian Matrices over the Octonions with Imaginary Eigenvalues*, arXiv:0902.2722v3 [math.RA] (2009).
- [7] J. C. Baez, *The Octonions*, Bull. Amer. Math. Soc., 39(2), 145-205 (2002).
- [8] J. Baez, *Inner Automorphisms of the Octonions*, The n-Category Café (Nov. 22, 2022). [\[Online\]](#).
- [9] A. Butter et al., *Extrapolating Jet Radiation with Autoregressive Transformers*, arXiv:2412.12074 [hep-ph] (2024).
- [10] A. Butter et al., *Simulation-Prior Independent Neural Unfolding Procedure*, arXiv:2507.15084 [hep-ph] (2025).